Do Now · retrieval

Lesson 076 · Algorithm Choice: Time, Space and Preconditions

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What must binary search assume?

Q2

Define auxiliary space.

Q3

Why can preprocessing change a choice?

Q4

Binary search in [1,3,5,7,9,11,13,15,17] for 15: give midpoint indexes.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Choose an algorithm based on input properties, particularly whether data are already sorted.

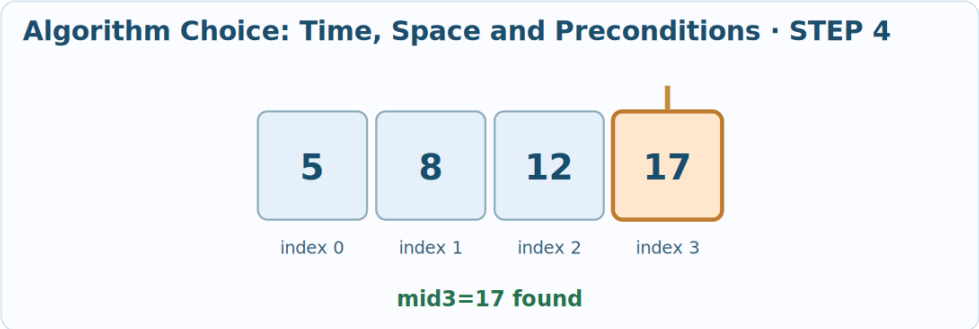
Core idea 2

Trace — Compare running time AND auxiliary memory: linear scan O(n) time, O(1) extra; iterative binary search O(log n), O(1) extra on sorted data.

Core idea 3

Explain — Sorting first takes time; consider that one-time cost and the number of future searches.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

precondition — Input requirement for correctness.

preprocessing — Preparing data before main operation.

amortise — Spread one-off cost across tasks.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

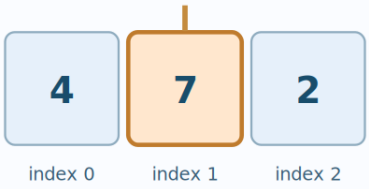
Choose a method to find 7 in unsorted [4,7,2].

Follow the worked method

1. Check the working — Linear search: inspect4 then7, returning index1.

Source model · final illustrated step

IDO · TRACE



Linear search: inspect4 then7, returning index1.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Check the working
mid3=9, mid5=16, mid4=12, return index4.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Choose linear/binary for unsorted [8,2,5,9], target5. [4]

Question 2

Trace binary search for 12 in [2,5,7,9,12,16,20]. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare one search in short unsorted list versus many searches in large fixed dataset. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Binary search must always be preferred even for one lookup in unsorted data.

Use key vocabulary from memory

1. Explain precondition in your own words.

2. Explain preprocessing in your own words.

3. Explain amortise in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.